COMPUTER VISION / MANUFACTURING QUALITY

Casting Defect Inspection

Technical assessment report

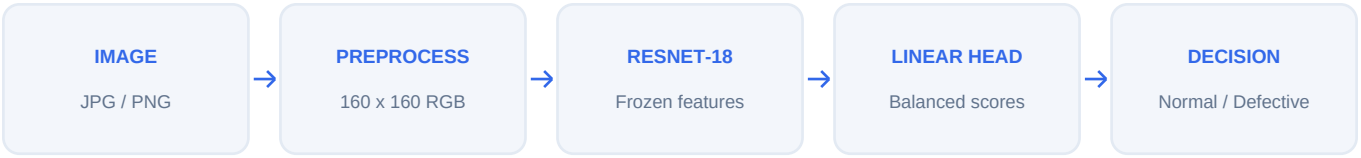
A visual inspection system for pump-impeller casting images. The trained model classifies each submitted image as **Normal** or **Defective** and returns a model score through a browser interface and HTTP API.

TEST ACCURACY	DEFECT PRECISION	DEFECT RECALL	DEFECT F1
98.60%	98.90%	98.90%	0.9890

System overview

Component	Implementation
Model	ImageNet-pretrained ResNet-18 with a frozen backbone and balanced logistic classification head.
Input / output	JPEG or PNG image; predicted class, class confidence, defect score, threshold and model version.
Evaluation	715 held-out test images; 705 correct classifications, 5 false positives and 5 false negatives.
Serving	FastAPI service with shared preprocessing, bounded uploads, structured errors, request IDs and health probes.

Inference architecture



INTERPRETATION

Results describe the supplied image-level benchmark. Pre-existing augmentation and missing physical-part IDs limit claims about independent factory products. Confidence scores are uncalibrated.

01 / DATA AND MODEL

Dataset preparation and model strategy

The original test set remains held out. Exact pixel deduplication precedes a stratified split of the supplied training pool.

Partition	Normal	Defective	Total
Training	2249	3006	5255
Validation	562	752	1314
Test	262	453	715

All 7,348 supplied 300 x 300 images decoded successfully. The audit excluded 64 exact duplicate training images, including any exact matches to test images. Remaining training data is split 80/20 with stratified labels and seed 42. The separate 512 x 512 source collection is excluded because its overlap with the augmented collection is uncertain.

Preprocessing and augmentation

Grayscale-looking RGB images are converted to RGB, resized as a whole to 160 x 160 and normalized with ImageNet channel statistics. No center crop is applied, preserving the outer rim. Each training image receives one additional mirrored view with a small positive or negative 8-degree rotation. Validation and test images are not augmented.

Architecture and imbalance

A frozen ResNet-18 produces 512 pooled features. Training-only feature standardization feeds a balanced, L2-regularized logistic head. Inverse-frequency class weights address the modest class imbalance. The scaler and learned coefficients are embedded in the saved Torch model, so inference does not require scikit-learn.

Validation-only selection

Setting	Selected strategy
Regularization	C in {0.001, 0.01, 0.1, 1, 10}; selected C = 0.1
Decision threshold	0.10 to 0.90 in steps of 0.02; selected threshold = 0.36
Objective	Maximum validation defect F2 subject to at least 95% precision; the constraint was met.
Test discipline	Model and threshold locked before test scoring; no refit on validation data.

DATA INDEPENDENCE

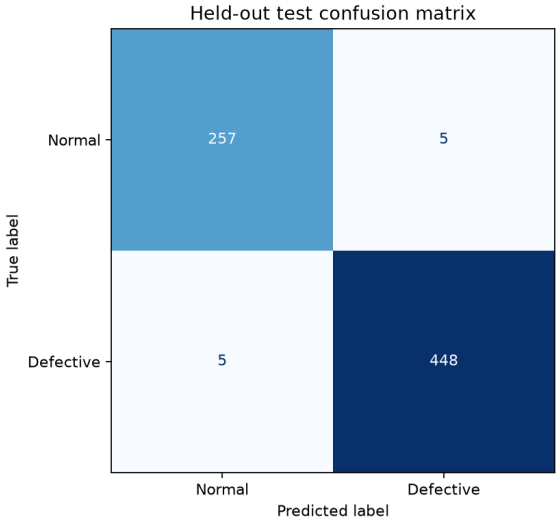
The audit flags 566/715 test images with a training dHash within four bits. Similar aligned castings can share hashes; these are screening candidates, not proven duplicates. Exact deduplication cannot establish physical-part independence.

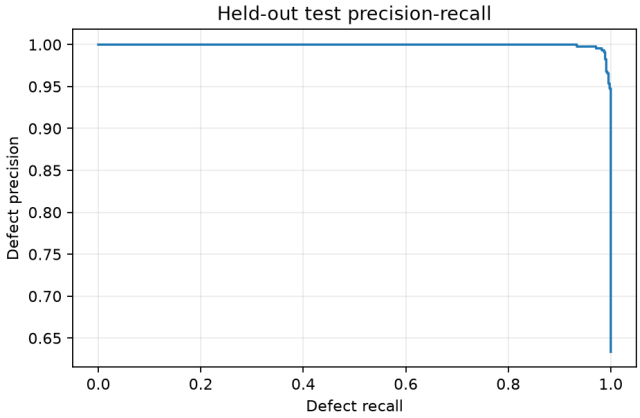
02 / EVALUATION

Held-out test results

Defective is the positive class. Classification uses the validation-selected threshold of 0.36; all values below come from the saved test predictions.

Metric	Test result	Interpretation
Accuracy	98.60%	705 of 715 images correctly classified
Defect precision	98.90%	448 true defects / 453 defect predictions
Defect recall	98.90%	448 detected / 453 labeled defects
Defect F1 / F2	0.9890 / 0.9890	Precision-recall operating-point summaries
ROC AUC	0.998964	Ranking performance across thresholds
Average precision	0.999411	Precision-recall curve summary
Majority baseline	63.36%	Always predict the majority test class





Confusion matrix: rows = true class; columns = predicted class. TN = 257, FP = 5, FN = 5, TP = 448.

Uncertainty and interpretation

Bootstrap 95% intervals are 97.62% to 99.44% for accuracy and 97.81% to 99.78% for defect recall. These intervals resample fixed image-level predictions; they do not resolve dependence between augmented views.

Full candidate metrics, individual predictions and class reports are retained in reports/metrics.json and reports/test_predictions.csv.

03 / ERROR ANALYSIS

False positives and false negatives

The test set contains five false positives and five false negatives. The examples below illustrate confident and localized errors. Observations are visual hypotheses, not operator-confirmed defect annotations.

FALSE POSITIVE

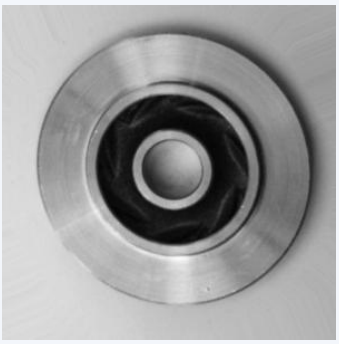


cast_ok_0_259.jpeg

Defect score: **0.912** | threshold: 0.36

Strong highlights on a Normal-labeled part; a confident false rejection.

FALSE POSITIVE

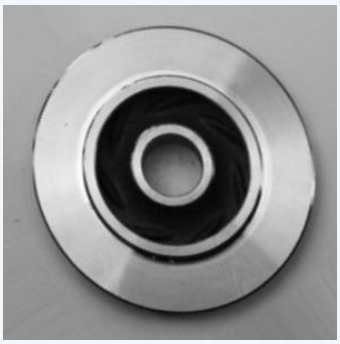


cast_ok_0_9926.jpeg

Defect score: **0.420** | threshold: 0.36

A near-threshold false rejection of a Normal-labeled part.

FALSE NEGATIVE



cast_def_0_2483.jpeg

Defect score: **0.054** | threshold: 0.36

Localized upper-rim roughness; the model still assigns a low defect score.

FALSE NEGATIVE



cast_def_0_3105.jpeg

Defect score: **0.067** | threshold: 0.36

Outer-rim irregularities on an otherwise regular-looking part are missed.

Operational consequences

False positives create unnecessary rejection or review of good parts. False negatives allow defective parts to pass. All five missed defects have scores below 0.18, so a narrow review band around the 0.36 threshold would not catch them.

Resizing can weaken tiny rim details; global pooled features may emphasize overall appearance. Lighting, surface texture and label ambiguity are plausible contributors. No observed test error was used to retune this submitted model. All ten cases are listed in reports/errors.json and discussed in docs/error_analysis.md.

04 / INFERENCE AND DEPLOYMENT

Service verification and operating limits

Capability	Implementation / evidence
Browser interface	Drag-and-drop image upload, whole-image preview, reference images, real predictions and JSON downloads.
Inference API	POST /predict; class, confidence, defect score, threshold, version, request ID and latency.
Input controls	JPEG/PNG only; 8 MiB file limit, bounded request body, minimum 32 x 32 pixels and maximum 16 megapixels.
Lifecycle and logging	Artifact checksum, startup warmup, readiness/liveness probes, request IDs and sanitized error responses.
Verification	16 API/inference checks and 12 desktop/mobile browser checks passed.
Optional Docker	Non-root user, read-only filesystem, CPU/memory limits and health check. Image build and health check passed; container inference not separately verified.

Measured latency

On the recorded Windows CPU environment, 50 sequential warm HTTP requests after five warmups measured **28.8 ms median** and **79.0 ms p95**. The service used two Torch threads and one active inference call. These figures describe localhost round-trip latency, not cold-start latency, container latency or a production throughput guarantee.

Known limitations

The classifier does not locate defects, identify defect types, inspect hidden surfaces or reject unrelated images as out of domain. Scores are uncalibrated. New lighting, geometry, motion blur and manufacturing batches fall outside the established evaluation. The supplied data cannot establish physical-part independence. Factory acceptance and production load testing are outside this verification scope.

Evidence and references

Model: artifacts/model.pt and artifacts/metadata.json. Dataset/splits: reports/dataset_analysis.json and reports/split_manifest.csv. HTTP evidence: reports/http_evidence.json. Test evidence: reports/test-results.xml and reports/ui-verification.json. Architecture: docs/architecture.md.

Dataset: Ravirajsinh Dabhi, casting product image data for quality inspection (Kaggle). Publisher acknowledgement: PILOT TECHNOCAST. Imagery retains CC BY-NC-ND 4.0 attribution.

Framework references: Torchvision ResNet-18; scikit-learn logistic regression; FastAPI image uploads.